

CS & IT ENGINEERING



Operating System

CPU Scheduling

Lecture -4

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Recap of Previous Lecture

doubt?



Topic

SRTF Scheduling

Topic

LF Scheduling

Topic

LRTF Scheduling

Topics to be Covered



Topic

Priority based Scheduling

Topic

Round Robin Scheduling



Topic : HRRN (Highest Response Ratio Next)

Process	Arrival Time	Burst Time
P1	0	6
P2	1	4
P3	5	1
P4	6	5



At time 6:-

$$RR(P2) = \frac{5+4}{4} = 2.25 \text{ (Highest)}$$

$$RR(P3) = \frac{1+1}{1} = 2$$

$$RR(P4) = \frac{0+5}{5} = 1$$

At time 10:-

$$RR(P3) = \frac{5+1}{1} = 6 \text{ (Highest)}$$

$$RR(P4) = \frac{4+5}{5} = 2.2$$



Topic : Priority Based Algorithm

Scheduling Criteria: highest priority process first | Tie breaker $\Rightarrow$ given in questⁿ

Type of Algorithm: Non-preemptive
or
preemptive





Topic : Priority Based Algorithm

Process	Arrival Time	Burst Time	Priority
P1	0	4	4
P2	1	2	5
P3	2	3	6
P4	3	1	10(Highest)
P5	4	2	9

Non-preemptive :-

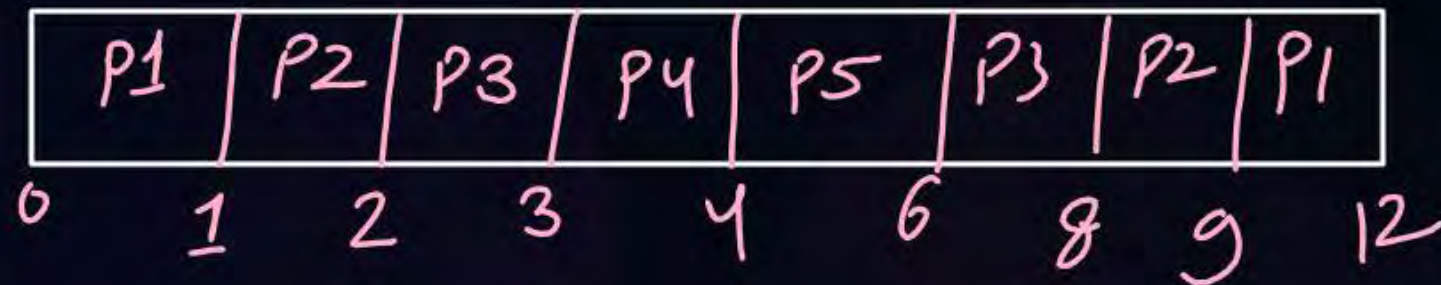




Topic : Priority Based Algorithm

Process	Arrival Time	Burst Time	Priority
P1	0	4	4
P2	1	2	5
P3	2	3	6
P4	3	1	10(Highest)
P5	4	2	9

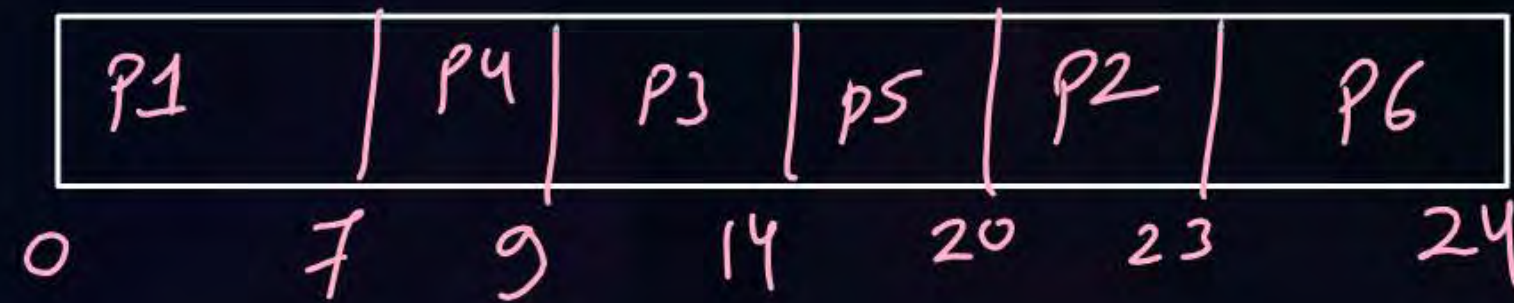
Preemptive :-





Topic : Priority Based Algorithm Question Non-Preemptive

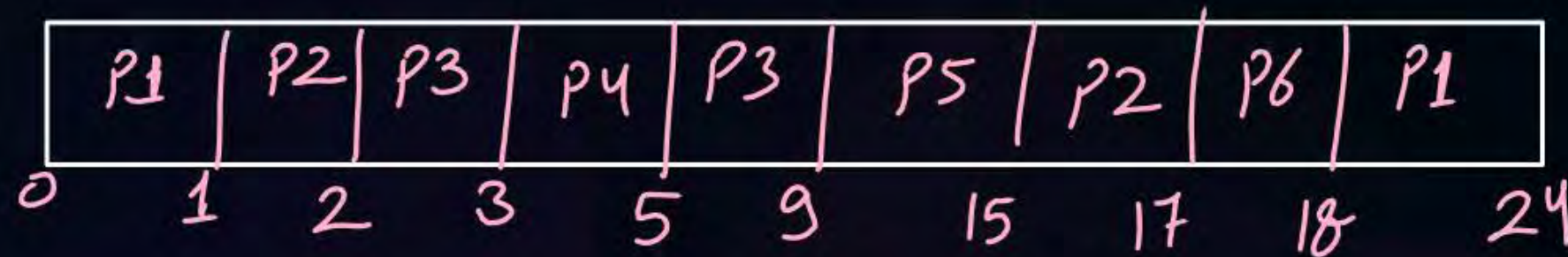
Process	Arrival Time	Burst Time	Priority
P1	0	7	9
P2	1	3	4
P3	2	5	2
P4	3	2	1 (Highest)
P5	4	6	3
P6	5	1	8





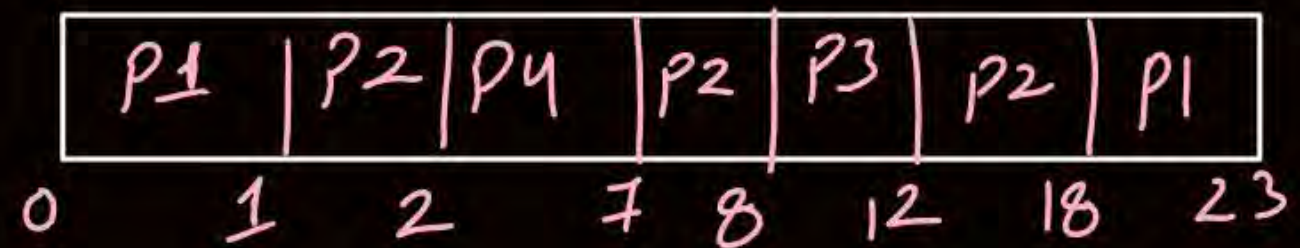
Topic : Priority Based Algorithm Question Preemptive

Process	Arrival Time	Burst Time	Priority
P1	0	7	9
P2	1	3	4
P3	2	5	2
P4	3	2	1 (Highest)
P5	4	6	3
P6	5	1	8



Ques) Preemptive

Process	AT	BT	Priority
P1	0	6	4
P2	1	8	5
P3	8	4	7 (Highest)
P4	2	5	6





Topic : Priority Based Algorithm



Advantages:

1. Better response for real time situations

Disadvantages:

2. Low Priority Processes may suffer from starvation



Priority

static

same priority for a process and it won't change.

dynamic

priority of processes may change

solution of starvation $\Rightarrow$ Aging

↑
increase priority of process when it waits for a certain time.



Topic : Round Robin (RR)

Scheduling Criteria: *Arrival time + Quantum time*

Type of Algorithm: *Preemptive*





Topic : Round Robin (RR)

$$Q = 2$$

Process	Arrival Time	Burst Time
P1	0	3
P2	0	6
P3	0	4
P4	0	5

Ready Queue

P1, P2, P3, P4

p1	p2	p3	p4	p1	p2	p3	p4	p2	p4	
0	2	4	6	8	9	11	13	15	17	18



Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time
P1	0	5
P2	1	4
P3	2	1
P4	3	3

P1	P2	P3	P1	P4	P2	P1	P4	
0	2	4	5	7	9	11	12	13

Time	Ready Queue
0	P1
2	P2 , P3, P1
4	P3, P1 , P4 , P2



Topic : Round Robin (RR)

$Q = 2$

Process	Arrival Time	Burst Time
P1	0	6
P2	1	5
P3	2	4
P4	3	3
P5	4	2
P6	5	4

Time	Ready Queue
0	P1
2	P2 , P3, P1
4	P3 , P1, P4, P5, P2
6	P1, P4, P5, P2, P6, P3 X X X X X X

no. of context switches
= 12

P1	P2	P3	P1	P4	P5	P2	P6	P3	P1	P4	P2	P6	
0	2	4	6	8	10	12	14	16	18	20	21	22	24

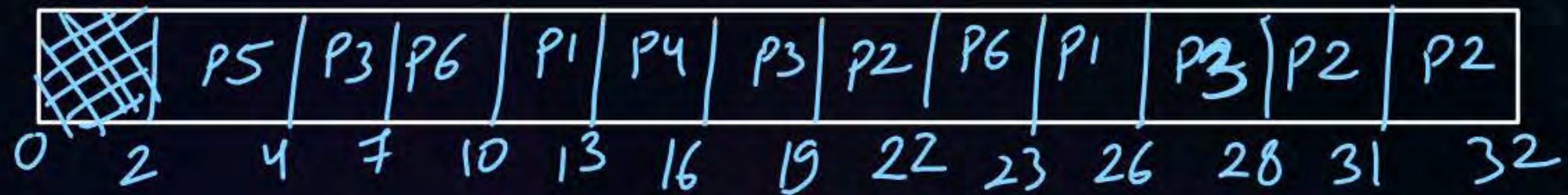


Topic : Round Robin (RR)

Q = 3

Process	Arrival Time	Burst Time
P1	5	6 3
P2	8	7 4 1
P3	3	8 5 2
P4	6	9
P5	2	2
P6	4	4 1

Time	Ready Queue
0	
2	P5
4	P3 P6
7	P6 P1, P4, P3
10	P1, P4, P3, P2, P6 Y X X Y





Topic : Round Robin (RR)

$Q = 3$

H.W.



Process	Arrival Time	Burst Time
P1	0	12
P2	0	5
P3	3	9
P4	5	6
P5	2	8
P6	4	2
P7	1	7



2 mins Summary

Topic

Priority based Scheduling

Topic

Round Robin Scheduling





Happy Learning

THANK - YOU

